

Algebra 1
Unit 2
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

1. The power of a power law of exponents states to keep the then _____ the powers.

☐ add
☐ subtract
☒ multiply

☒ base
☐ coefficient
☐ exponent

and

2. Diana simplified the algebraic expression and made an error. Find and fix the mistake.

$$\begin{aligned} &(-3x^3y^5)^3(1.5xy^0) \\ &\quad (-3x^9y^{15})(1.5x) \\ &(-3y \cdot 1.5)(y^{15})(x^9 \cdot x) \\ &\quad -4.5x^9y^{15} \end{aligned}$$

$$\begin{aligned} &(-3x^3y^5)^3(1.5xy^0) \\ &\quad (-27x^9y^{15})(1.5x) \\ &\quad -40.5x^{10}y^{15} \end{aligned}$$

3. Write a note to Diana explaining the error she made in the previous question. Remember Diana, the power of a power law says to multiply all powers. This includes the leading coefficient. In the term $(-3x^3y^5)^3$, -3 has a power of 1. So, the power of 3 must be multiplied to the power of 1 also. In the term x , there is always a power of 1. When there is none, there is one, x^1 .

Determine an equivalent algebraic monomial expression for each expression using the laws of exponents. Show your work.

4. $(-8a^5b)(3ab^4)$

$$\begin{aligned} &(-8 \cdot 3)(a^5 \cdot a)(b \cdot b^4) \\ &-24a^6b^5 \end{aligned}$$

5. $(\frac{-1}{4}x^3y^2)(4x^8y^3)(-5y^2)$

$$\begin{aligned} &(\frac{-1}{4} \cdot 4 \cdot -5)(x^3 \cdot x^8)(y^2 \cdot y^3 \cdot y^2) \\ &5x^{11}y^7 \end{aligned}$$

$$6. \quad (2.8kj^9)(3.5k^8)(\frac{1}{3}k^5j^2)$$

$$(2.8 \cdot 3.5 \cdot \frac{1}{3})(k \cdot k^8 \cdot k)(j^9 \cdot j^2)$$

$$\frac{49}{15}k^{14}j^{11} \quad \text{or} \quad 3.2\bar{6}k^{14}j^{11}$$

$$7. \quad (-3mn^5p^4)^3$$

$$-3^3m^3n^{15}p^{12}$$

$$-27m^3n^{15}p^{12}$$

$$8. \quad (2.3c^4d)(-2.4c^{-3}d^0)(-c^2d^{-5})$$

$$(2.3 \cdot -2.4 \cdot -1)(c^4 \cdot c^{-3} \cdot c^2)(d \cdot d^0 \cdot d^{-5})$$

$$\frac{5.52c^3}{d^4}$$

$$9. \quad (\frac{5}{6}r^{-8})^{-2}(-9.6r^5s^4)^0$$

$$(\frac{5}{6})^{-2}(r^{-8})^{-2} \cdot 1$$

$$(\frac{6}{5})^2 \cdot r^{16}$$

$$\frac{36}{25}r^{16} \quad \text{or} \quad 1.44r^{16}$$

$$10. \quad (-8y^5z)^3(5y^7z^6)^3$$

$$-8^3y^{15}z^3 \cdot 5^3y^{21}z^{18}$$

$$-512y^{15}z^3 \cdot 125y^{21}z^{18}$$

$$-512 \cdot 125y^{36}z^{21}$$

$$-64,000y^{36}z^{21}$$